Module V: Open-Ended Projects in LaTeX

Study Notes · MAT5FS112 · Installation, Overleaf, chapters, slides, posters, transliteration

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1. Installing LaTeX

You need a *distribution* (the engine plus packages) and an editor. The usual choices are:

- **TeX Live** (Linux, macOS, Windows). On Arch Linux, install `texlive-basic texlive-latex texlive-latexrecommended texlive-latexextra texlive-fontsrecommended texlive-bibtexextra` with `pacman`. On Debian or Ubuntu: `texlive-full` (large, but complete).
- **MiKTeX** (Windows), which installs missing packages on demand.
- **MacTeX** (macOS).

After installation, check the tools are on the path:

```
pdflatex --version
bibtex --version
makeindex --version
```

A good editor (TeXstudio, TeXworks, VS Code with LaTeX Workshop, or Vim with a LaTeX plugin) builds the document and shows errors.

2. Overleaf

Overleaf (`overleaf.com`) is an online editor. It needs no installation, has collaborative editing and compiles the document on its servers. The steps are:

1. Create a free account and choose *New Project*, then *Blank Project*.
2. Paste the code from any file in these notes into `main.tex`.
3. Press *Recompile*. The PDF appears on the right, and the compile log is under *Logs and output*.
4. Use the menu *Menu* → *Settings* to choose the compiler (pdfLaTeX, XeLaTeX or LuaLaTeX) and the main file.

> **Keep a local copy.** Keep a local copy of your `.tex` file. Overleaf can be unavailable, and the same code compiles on a local TeX Live.

3. Writing a chapter of a book

A book uses the `report` class (or `book`), where the top level is `\chapter`. The skeleton below is a template for a chapter of a textbook with theorems and equations:

```
\documentclass[11pt,a4paper]{report}
\usepackage{amsmath,amsthm,amssymb}
\newtheorem{theorem}{Theorem}[chapter]
\begin{document}
\chapter{Introduction to Sets}
\section{Basic notions}
\begin{theorem}
...
\end{theorem}
\end{document}
```

Here is the same structure, compiled, in this `article` file.

> **Theorem 1 (Sample result)**
>
> Every finite set has finitely many subsets, namely 2^n for a set with n elements.

Proof.

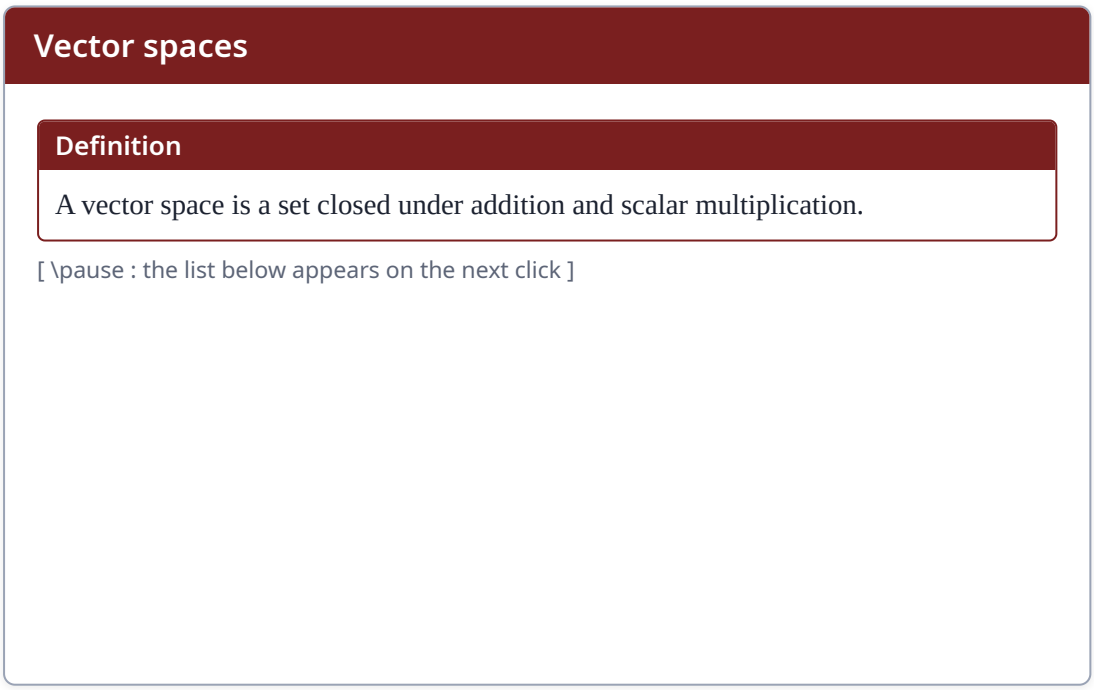
Each element is either in a subset or not, which gives two choices per element and 2^n choices in all. ∎

4. Slides with Beamer

`beamer` is a document class for presentations. Each slide is a `\frame`. The code below belongs in a file of its own that starts with `\documentclass{beamer}`; it is shown here as a listing LISTING ONLY:

```
\documentclass{beamer}
\usetheme{Madrid}
\usecolortheme{beaver}
\title{Linear Algebra Basics}
\author{Your Name}
\date{\today}
\begin{document}
\frame{\titlepage}
\begin{frame}{Vector spaces}
  \begin{block}{Definition}
    A vector space is a set closed under addition and scalar multiplication.
  \end{block}
  \pause
  \begin{itemize}
    \item Eight axioms must hold.
    \item Examples: $\mathbb{R}^n$, polynomials.
  \end{itemize}
\end{frame}
\end{document}
```

A mock-up of the `Vector spaces` frame, after the first step of `\pause`:



Key ideas:

- `\frame{...}` or `\begin{frame}{Title}...\end{frame}`.
- `\pause` reveals the content in steps.
- `\begin{block}{...}` gives a highlighted box.

5. Posters

A poster is a single large page with several columns. The `beamerposter` package is the usual tool. Without it, a poster-like layout can be built with `multicols` and a `minipage`. Here is a compiled version of a two-column poster section:

```
\begin{multicols}{2}
\section*{Motivation}
Matrices describe linear maps. Their determinant measures how the
map scales area:
\[
  \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.
\]

\section*{Result}
A matrix is invertible if and only if its determinant is non-zero.
\columnbreak
\section*{Example}
For $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\det A = 5 \neq 0$,
so $A$ is invertible.
\end{multicols}
```

Motivation

Matrices describe linear maps. Their determinant measures how the map scales area:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Result

A matrix is invertible if and only if its determinant is non-zero.

Example

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\det A = 5 \neq 0$, so A is invertible.

6. Transliteration of Indian languages

Transliteration writes an Indian script in Latin letters. Scholarly Sanskrit uses IAST (International Alphabet of Sanskrit Transliteration), whose letters need diacritics. In `pdflatex` these are made with accent commands. This is compiled in the source file:

```
\begin{center}
\begin{tabular}{ll}
\toprule
Command & Letter \\
\midrule
\verb|\={a}| & $\bar{a}$ as in \emph{R\={a}ma} \\
\verb|\'{s}| & \'{s} as in \emph{\'{s}iva} \\
\verb|\d{t}| & \d{t} as in \emph{\d{t}ha} \\
\verb|\d{n}| & \d{n} as in \emph{Ka\d{n}a} \\
\verb|\.{m}| & \.{m} as in \emph{Sa\.{m}sk\d{r}ta} \\
\verb|\d{r}| & \d{r} as in \emph{\d{r}\d{s}i} \\
\bottomrule
\end{tabular}
\end{center}
```

COMMAND	LETTER	EXAMPLE
`\={a}`	ā	as in *Rāma*
`\'{s}`	ś	as in *śiva*
`\d{t}`	ṭ	as in *ṭha*
`\d{n}`	ṇ	as in *Kaṇa*
`\.{m}`	ṁ	as in *Saṁskṛta*
`\d{r}`	ṛ	as in *ṛṣi*

Example in running text: *Rāmāyaṇa* and *Bhagavad-gītā*.

6.1 Devanagari and Malayalam (XeLaTeX or LuaLaTeX)

Printing the script itself needs a Unicode engine and a font that has the glyphs. The usual setup is `fontspec` with `polyglossia`. This listing is *not* compiled in these notes, because it needs XeLaTeX and an installed font. Replace each placeholder with the word typed in a Unicode-capable editor: LISTING ONLY

```
\documentclass{article}
\usepackage{fontspec}
\usepackage{polyglossia}
\setdefaultlanguage{hindi}              % or sanskrit, malayalam
\newfontfamily\devanagarifont{Noto Serif Devanagari}
\newfontfamily\malayalamfont{Noto Serif Malayalam}
\begin{document}
{\devanagarifont <Devanagari word>} \quad {\malayalamfont <Malayalam word>}
\end{document}
```

Compile with `xelatex file.tex` (or `lualatex`). Check the font names with `fc-list` first, because they must match the installed font exactly.

7. Checklist for an open-ended project

1. Set up the distribution and the editor, and compile a test file.
2. Choose the document class (`article`, `report` or `beamer`).
3. Add packages only as needed, and keep the preamble short.
4. Compile twice for references, the table of contents and the list of figures.
5. Run BibTeX or makeindex when the document uses them.
6. Read the log: the first error line is the one to fix.